

B rainstorming & Idea Prioritiz ation

This document records the structured brainstorming sessions conducted to identify, evaluate, and prioritize the core ideas that form the foundation of OptiCrop.

1. Problem Space Exploration

Team members individually generated ideas around the central challenge: *How can machine learning help smallholder farmers choose the right crop for their land?*

Key Problems Identified

- " Farmers rely on experience and intuition, leading to sub-optimal crop selection
- " Soil nutrient data (N, P, K) is rarely translated into actionable planting advice
- " Climate variability (temperature, humidity, rainfall) is not factored into decisions
- " Existing tools are expensive, complex, or inaccessible to rural communities
- " Crop failure leads to economic loss and food insecurity in developing regions

2. B rainstormed Solutions

- Idea #1** Rule-based expert system using agronomist-defined thresholds for each crop
- Idea #2** Mobile app with GPS-based weather data and historical crop yield mapping
- Idea #3** ML-powered prediction engine trained on multi-parameter soil and climate datasets
- Idea #4** Community forum platform where farmers share crop success stories
- Idea #5** IoT soil sensor network feeding a real-time prediction dashboard
- Idea #6** SMS-based advisory service for areas with limited internet connectivity

3. Idea Prioritiz ation Matrix

Each idea was evaluated on four criteria: Technical Feasibility (TF), Impact (I), Novelty (N), and Time-to-Build (TTB). Scores are 1-5; higher is better.

Idea	TF	Impa ct	Nove lty	TT B	Total	Priority
Rule-based expert system	3	3	2	4	12	Low
Mobile app + GPS weather	3	4	3	3	13	Medium
ML prediction engine (chosen)	5	5	5	4	19	HIGH
Community forum platform	4	2	2	5	13	Medium
IoT sensor network	2	5	5	1	13	Deferred
SMS advisory service	4	4	2	4	14	Medium

4. Selected Idea: ML -Powered Crop Recommendation

The Machine Learning Prediction Engine scored highest on all dimensions. It offers the best balance of real-world impact, technical novelty, and feasibility within the project timeline. The core solution, OptiCrop, will:

- " Accept seven soil/climate parameters as input (N, P, K, temp, humidity, pH, rainfall)
- " Use trained ML classifiers to recommend the optimal crop
- " Expose recommendations via a responsive Flask web application
- " Allow future extension to IoT sensors and mobile SMS delivery

The prioritization exercise also highlighted two complementary features model explainability (SHAP values) and multilingual UI that will be added in the roadmap as Phase 2 enhancements.